

```
In [1]: ▶ import numpy as np
import cv2
```

```
In [2]: ▶ cam=cv2.VideoCapture(0)
print("width: ",cam.get(3))
```

width: 640.0

```
In [3]: ▶ print("width: ",cam.get(4))
```

width: 480.0

```
In [4]: ▶ import matplotlib.pyplot as plt
```

```
-----
ModuleNotFoundError                                Traceback (most recent call last)
<ipython-input-4-c74fbadd359f> in <module>
----> 1 import matplotlib.pyplot as plt

ModuleNotFoundError: No module named 'matplotlib'
```

```
In [5]: ▶ img = np.zeros((512,512,3),np.uint8)+255
### (img,start,end,color(rgb),thickness)
cv2.line(img,(100,100),(300,400),(255,0,0),3)
cv2.line(img,(300,100),(100,400),(255,0,0),3)
plt.imshow(img)
plt.show()
```

```
-----
NameError                                           Traceback (most recent call last)
<ipython-input-5-e98bcc2975a8> in <module>
      3 cv2.line(img,(100,100),(300,400),(255,0,0),3)
      4 cv2.line(img,(300,100),(100,400),(255,0,0),3)
----> 5 plt.imshow(img)
      6 plt.show()

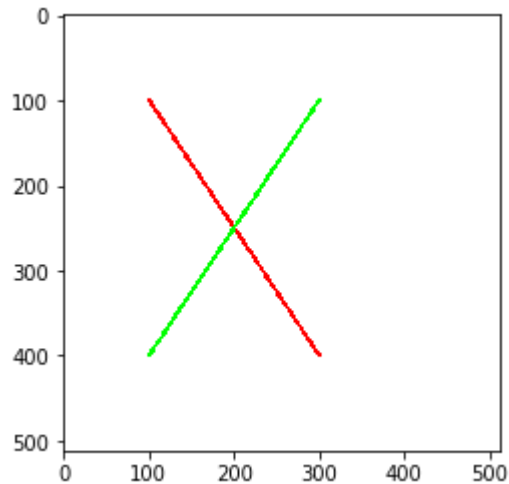
NameError: name 'plt' is not defined
```

```
In [6]: ▶ import cv2
```

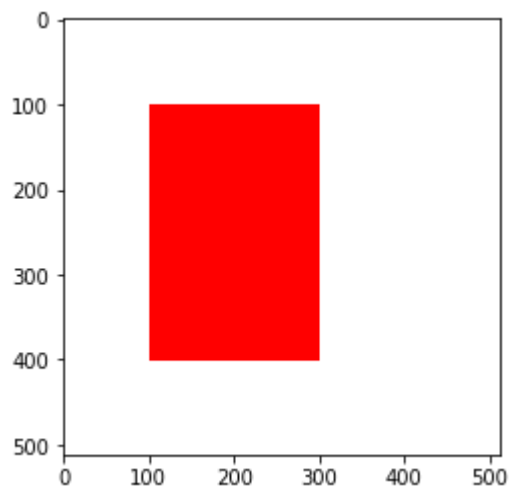
```
In [7]: ▶ import numpy as np
```

```
In [8]: ▶ import matplotlib.pyplot as plt
```

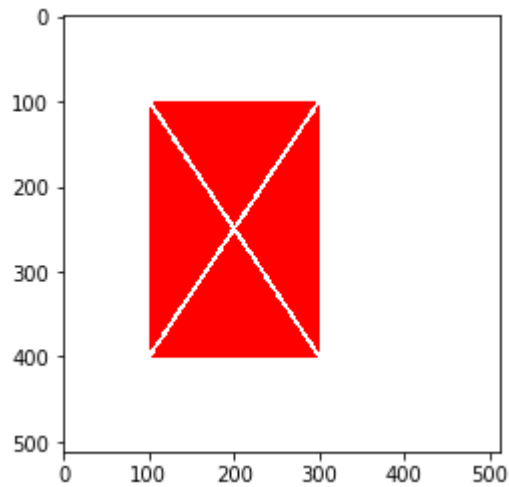
```
In [10]:  img = np.zeros((512,512,3),np.uint8)+255
        ### (img,start,end,color(rgb),thickness)
        cv2.line(img,(100,100),(300,400),(255,0,0),3)
        cv2.line(img,(300,100),(100,400),(0,255,0),3)
        plt.imshow(img)
        plt.show()
```



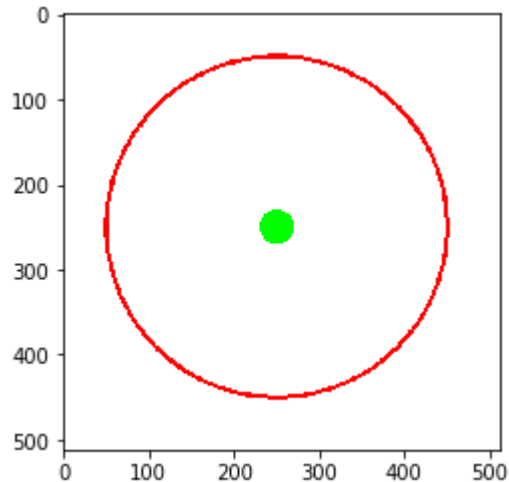
```
In [12]:  img = np.zeros((512,512,3),np.uint8)+255
        ### (img,start,end,color(rgb),thickness)
        cv2.rectangle(img,(100,100),(300,400),(255,0,0),-1)
        plt.imshow(img)
        plt.show()
```



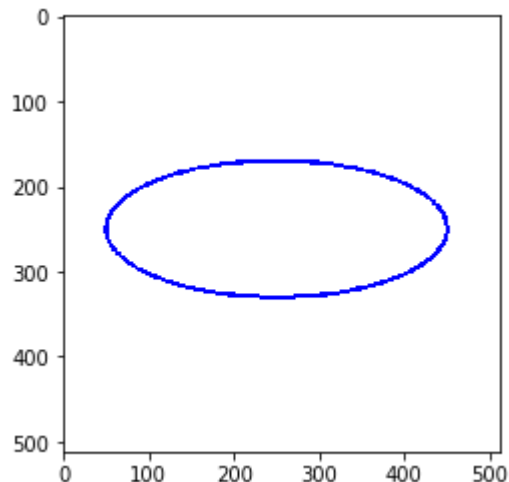
```
In [13]:  img = np.zeros((512,512,3),np.uint8)+255
### (img,start,end,color(rgb),thickness)
cv2.rectangle(img,(100,100),(300,400),(255,0,0),-1)
cv2.line(img,(100,100),(300,400),(255,255,255),3)
cv2.line(img,(300,100),(100,400),(255,255,255),3)
plt.imshow(img)
plt.show()
```



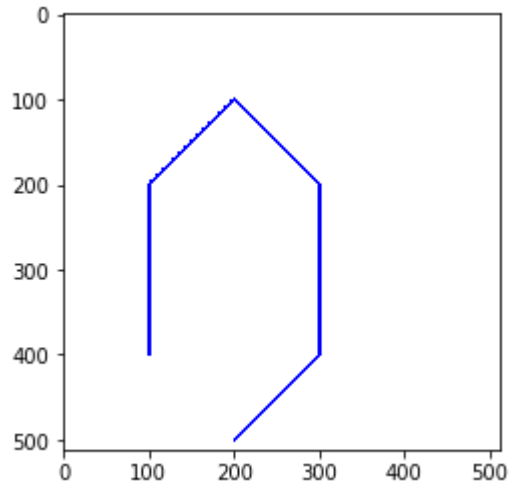
```
In [15]:  img = np.zeros((512,512,3),np.uint8)+255  
### (img,center,radius,color(rgb),thickness)  
cv2.circle(img,(250,250),200,(255,0,0),3)  
cv2.circle(img,(250,250),20,(0,255,0),-1)  
plt.imshow(img)  
plt.show()
```



```
In [16]:  img = np.zeros((512,512,3),np.uint8)+255  
### (img,center,(major,minor),angle,start angle,end angle,color(rgb),thickness)  
cv2.ellipse(img,(250,250),(200,80),0,0,360,(0,0,255),3)  
plt.imshow(img)  
plt.show()
```

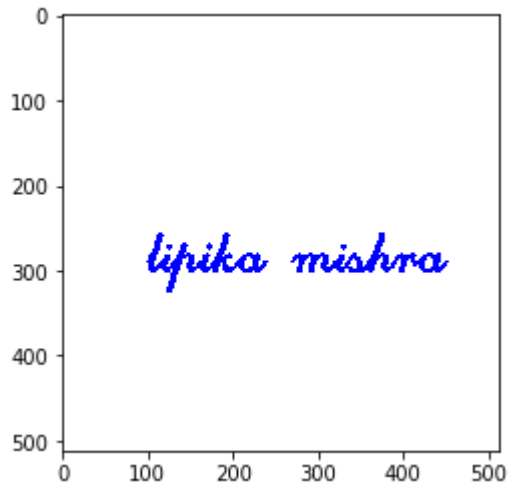


```
In [20]:  img = np.zeros((512,512,3),np.uint8)+255
        ### (img,[points],isclosed,color(rgb),thickness)
        points = np.array([[100,400],[100,200],[200,100],[300,200],[300,400],[200,500]])
        cv2.polylines(img,[points],False,(0,0,255),4)
        plt.imshow(img)
        plt.show()
```

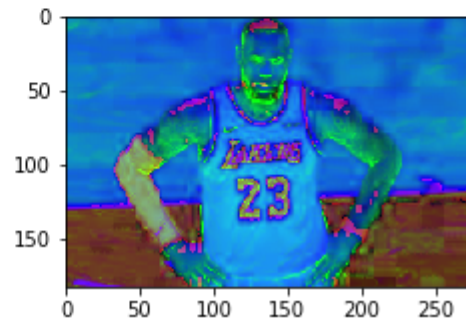
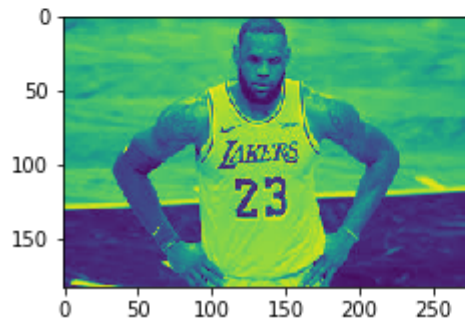
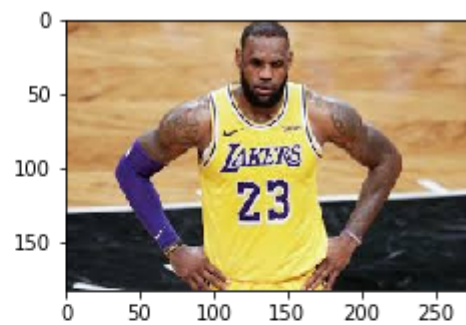
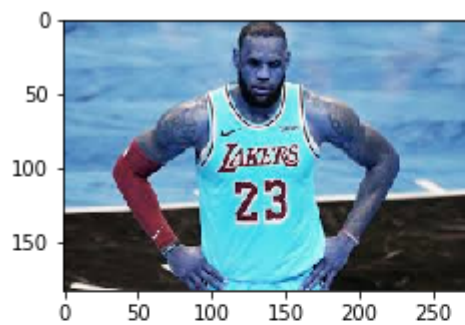


```
In [24]: ▶ img = np.zeros((512,512,3),np.uint8)+255
        ### (img,text,position,fontstyle,fontsize,color(rgb),boldness)
        name=input("your name please")
        cv2.putText(img,name,(100,300),cv2.FONT_HERSHEY_SCRIPT_COMPLEX,2,(0,0,255),3)
        plt.imshow(img)
        plt.show()
```

your name please
lipika mishra



```
In [25]: path = r"C:\Users\KIIT\Desktop\LeBron_James.jpeg"
img = cv2.imread(path)
rgb = cv2.cvtColor(img,cv2.COLOR_BGR2RGB)
gray = cv2.imread(path,0)
hsv = cv2.cvtColor(img,cv2.COLOR_BGR2HSV)
plt.figure(figsize=(8,8))
plt.subplot(2,2,1)
plt.imshow(img)
plt.subplot(2,2,2)
plt.imshow(rgb)
plt.subplot(2,2,3)
plt.imshow(gray)
plt.subplot(2,2,4)
plt.imshow(hsv)
plt.show()
```

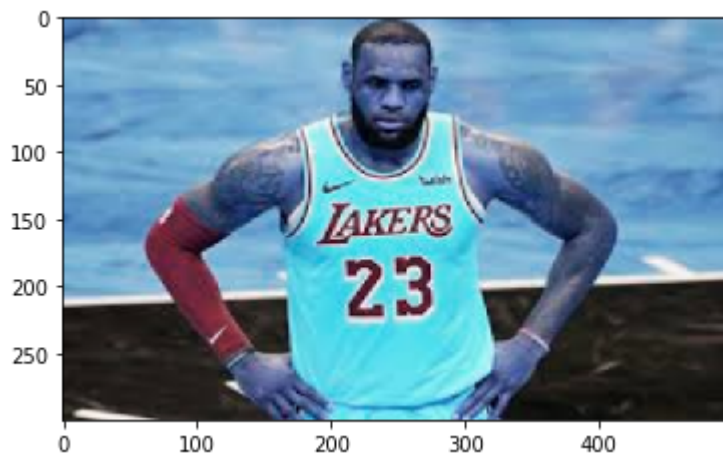


```
In [26]: path = r"C:\Users\KIIT\Desktop\LeBron_James.jpeg"
img = cv2.imread(path)
rgb = cv2.cvtColor(img,cv2.COLOR_BGR2RGB)
gray = cv2.imread(path,0)
hsv = cv2.cvtColor(img,cv2.COLOR_BGR2HSV)
plt.figure(figsize=(15,6))
labels = ["BGR Image","RGB Image","Grayscale","HSV Image"]
images = [img,rgb,gray,hsv]
for i in range(len(images)):
    plt.subplot(1,4,i+1)
    plt.title(labels[i])
    plt.xticks([])
    plt.yticks([])
    if labels[i] == "Grayscale":
        plt.imshow(images[i],cmap="gray")
    else:
        plt.imshow(images[i])
plt.show()
```

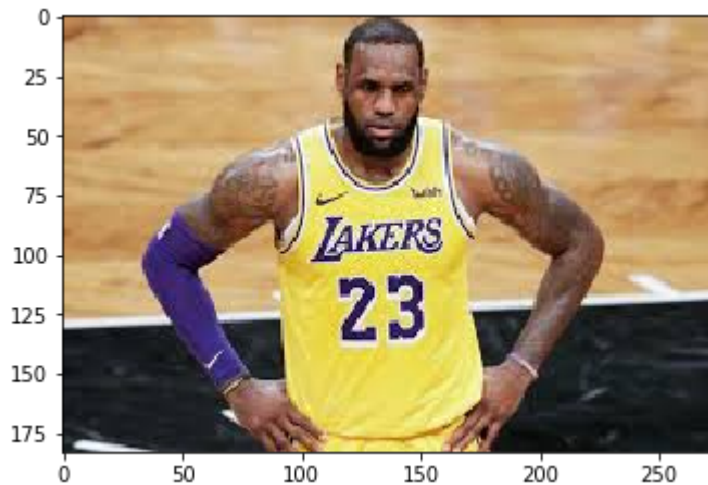


```
In [27]: path = r"C:\Users\KIIT\Desktop\LeBron_James.jpeg"
new_path = r"C:\Users\KIIT\Desktop\LeBron_James_new.jpg"
img = cv2.imread(path)
rgb = cv2.cvtColor(img,cv2.COLOR_BGR2RGB)
new = cv2.resize(img,(500,300)) ## (width,height)
cv2.imwrite(new_path,new) ## save a image as a different file
plt.imshow(new)
```

Out[27]: <matplotlib.image.AxesImage at 0x2215d1f96c8>



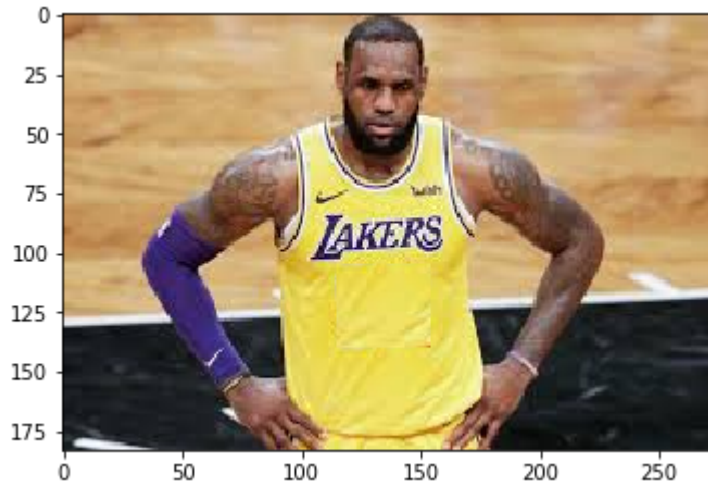

```
In [28]: ▶ path = r"C:\Users\KIIT\Desktop\LeBron_James.jpeg"
img = cv2.imread(path)
rgb = cv2.cvtColor(img,cv2.COLOR_BGR2RGB)
text = rgb[80:105,105:160]
num = rgb[105:140,115:155]
plt.imshow(rgb)
plt.show()
```



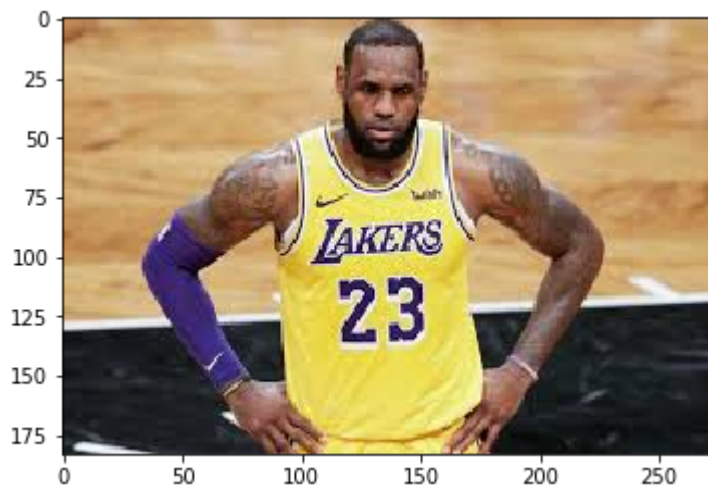
```
In [29]: ▶ path = r"C:\Users\KIIT\Desktop\LeBron_James.jpeg"
img = cv2.imread(path)
rgb = cv2.cvtColor(img,cv2.COLOR_BGR2RGB)
text = rgb[80:105,105:160]
num = rgb[105:140,115:155]
plt.imshow(rgb)
plt.show()
```



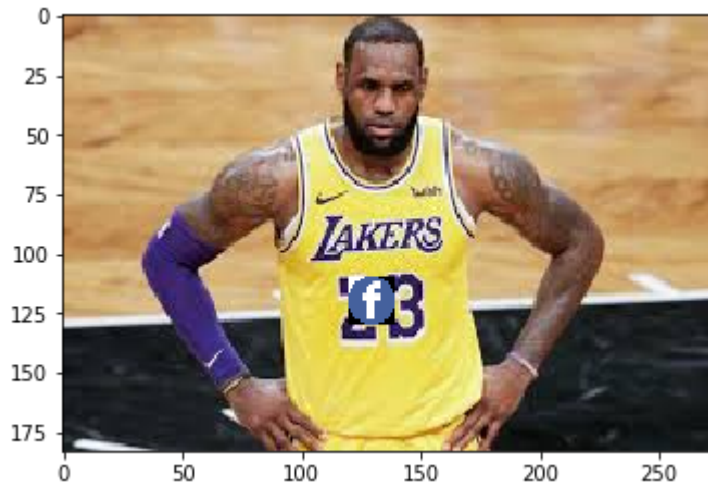
```
In [30]: path = r"C:\Users\KIIT\Desktop\LeBron_James.jpeg"
img = cv2.imread(path)
rgb = cv2.cvtColor(img,cv2.COLOR_BGR2RGB)
text = rgb[80:105,105:160]
num = rgb[105:140,115:155]
rgb[105:140,115:155] = rgb[140:175,115:155]
plt.imshow(rgb)
plt.show()
```



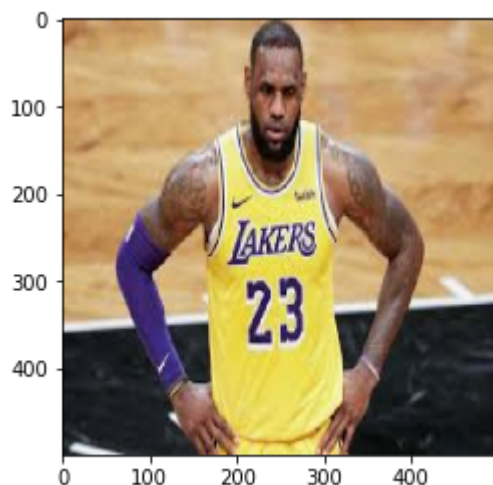
```
In [31]: path = r"C:\Users\KIIT\Desktop\LeBron_James.jpeg"
img = cv2.imread(path)
rgb = cv2.cvtColor(img,cv2.COLOR_BGR2RGB)
text = rgb[80:105,105:160]
num = rgb[105:140,115:155]
plt.imshow(rgb)
plt.show()
```



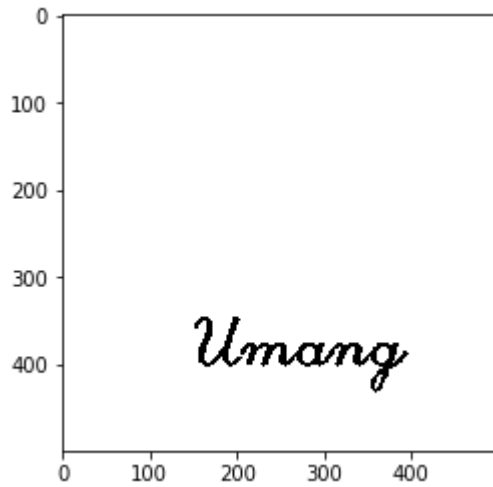
```
In [34]: icon_path = r"C:\Users\KIIT\Desktop\facebook_f.png"
icon = cv2.imread(icon_path)
rgb_icon = cv2.cvtColor(icon,cv2.COLOR_BGR2RGB)
rgb_icon = cv2.resize(rgb_icon,(20,20))
rgb[110:130,120:140] = rgb_icon
plt.imshow(rgb)
plt.show()
```



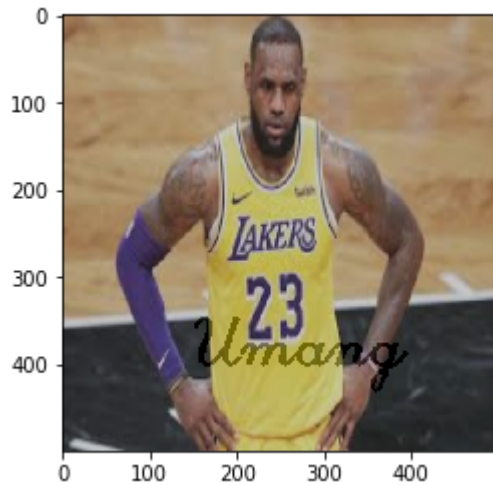
```
In [35]: path = r"C:\Users\KIIT\Desktop\LeBron_James.jpeg"
img = cv2.imread(path)
rgb = cv2.cvtColor(img,cv2.COLOR_BGR2RGB)
rgb = cv2.resize(rgb,(500,500))
plt.imshow(rgb)
plt.show()
```



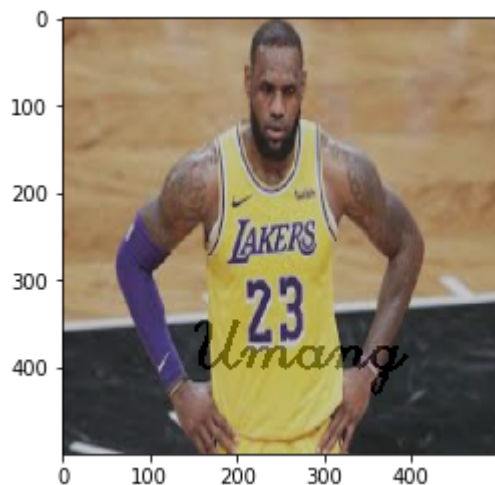
```
In [36]: ▶ name = np.zeros((500,500,3),np.uint8)+255  
cv2.putText(name,"Umang",(150,400),cv2.FONT_HERSHEY_SCRIPT_COMPLEX,2.5,(0,0,0)  
plt.imshow(name)  
plt.show()
```



```
In [37]: ▶ output = cv2.addWeighted(rgb,0.7,name,0.3,-50)  
plt.imshow(output)  
plt.show()
```



```
In [38]: output = cv2.addWeighted(rgb,0.7,name,0.3,-50)
plt.imshow(output)
plt.show()
```



```
In [39]: name = np.zeros((500,500,3),np.uint8)+255
icon_path = r"D:\Desktop\Open CV\icon.png"
icon = cv2.imread(icon_path)
rgb_icon = cv2.cvtColor(icon,cv2.COLOR_BGR2RGB)
rgb_icon = cv2.resize(rgb_icon,(100,100))
name[350:450,200:300] = rgb_icon
plt.imshow(name)
plt.show()
```

error Traceback (most recent call last)

```
<ipython-input-39-abf6d95ea3fe> in <module>
      2 icon_path = r"D:\Desktop\Open CV\icon.png"
      3 icon = cv2.imread(icon_path)
----> 4 rgb_icon = cv2.cvtColor(icon,cv2.COLOR_BGR2RGB)
      5 rgb_icon = cv2.resize(rgb_icon,(100,100))
      6 name[350:450,200:300] = rgb_icon
```

error: OpenCV(4.5.3) C:\Users\runneradmin\AppData\Local\Temp\pip-req-build-c213r8zm\opencv\modules\imgproc\src\color.cpp:182: error: (-215:Assertion failed) !_src.empty() in function 'cv::cvtColor'

In []: ▶